

1. Messaging

The message of this narrative visualization is that a country's population age is one of the strongest predictors of how high the fatality rate of COVID-19 was for that country. Older populations experienced significantly higher death rates than younger ones, both across individual countries and on a global scale. The narrative visualization's scenes walk the user from the broad global trend (country by country) to a direct quantification of that pattern (roughly 12x higher death rate for the oldest age bucket versus the youngest), then lastly lets the user explore the underlying granular data themselves with interactive filtering and sorting.

2. Narrative Structure

This narrative visualization follows the martini glass structure and has 3 scenes. Scenes 1 and 2 are a guided narrative with no user-driven filtering or interaction. The user just watches the data reveal itself in buckets and reads the final annotations that make the point. Scene 3 lets the user interact through filtering and sorting. The user gets a year-range slider and a country dropdown, and can also click directly on the map to view/pin a country's details under the legend. This allows the user to explore any country and year combination on their own instead of following a fixed path like in scenes 1 and 2.

3. Visual Structure

All three scenes share the same margins, the same four-color median age-group scale (light yellow to dark red, corresponding to youngest to oldest), and the same d3 annotation style, so the narrative visualization reads as one connected story rather than three separate charts. Scene 1 contains a log-log scatter plot of deaths vs. cases per country, with dots colored by median-age group. The dots fade in one age group at a time, youngest to oldest, this clearly shows the trend (older population, higher fatality quadrant) as it forms, rather than dumping all 189 points on the viewer at once and letting them decipher the trend themselves. Scene 2 contains a bar chart of average death rate by age group. The bars grow up from the x-axis in the same youngest-to-oldest order and each is labeled with its exact numerical value once it finishes growing. This turns the same color progression into a literal step-by-step visual bar size growth. Scene 3 contains a choropleth world map using the identical color scale, so a viewer who just spent two scenes learning that dark red means an older population and a high death rate can immediately read the map having to learn a new legend. The scene 3 chart legend is in the same position (top-right margin) as every scene. The chart legend is the visual anchor that carries the consistent color meaning across all scenes.

4. Scenes

Scene 1 - Global Fatality Trend: a scatterplot of every country's average cases and deaths per 100k from 2020 to 2023, colored by median-age group, with two annotations drawing attention to Bulgaria (oldest, in the highest fatality rate upper right quadrant) and Niger (youngest, in the lowest fatality rate lower left quadrant).

Scene 2 - Age Drives Mortality: a bar chart of the average death rate for each age group, with an annotation quantifying and drawing attention to the approximately 12x gap between the oldest and youngest median age buckets.

Scene 3 - Explore the World: an interactive choropleth world map with a year-range slider and a country dropdown, where clicking a country directly on the map displays/pins the exact same detail panel as selecting it from the dropdown.

The scenes are ordered 1 to 3 to achieve separate and distinct goals. Scene 1 shows the global trend by displaying a datapoint for every country and the trend for older countries to be located in higher fatality rate quadrants. Scene 2 collapses that same data into one number per age group so the correlation is extremely visual perceptible by the significant size difference of the bars. Scene 3 then hands the data to the viewer so they can check the narrative's claim against any country or year range themselves.

5. Annotations

Scene 1 uses 2 d3 annotations to draw attention to Bulgaria (oldest, in the highest fatality rate upper right quadrant) and Niger (youngest, in the lowest fatality rate lower left quadrant). This is so the user takes note of how different the fatality rates are for these two countries and how it relates to their position in the scatterplot. Scene 2 uses 1 d3 annotation to draw attention to the last bar to appear, the bar for the oldest median age group. This is to draw attention to the significant size and numerical value difference of the oldest and youngest median age buckets. The annotations do change within a single scene, in both Scenes 1 and 2 the annotations start invisible and only fade in after every dot in Scene 1 or bar in Scene 2 has finished its reveal animation. This is so the annotations deliver their conclusions after the full buildup of evidence.

6. Parameters

This narrative visualization has four parameters that make up its state object. The current scene, the start year, the end year, and the selected/pinned country. These are the only variables that determine what gets drawn on the narrative visualization. Every scene-drawing function reads its inputs directly from this state object, so the parameters directly control scene construction. Scenes 1 and 2 don't read anything except for the scene number. They always aggregate the full 2020 to 2023 year time range and country is irrelevant to them. Scene 3 is the only scene that uses the other parameters. The start/end year parameters determine which rows get filtered and averaged into each country's color and stats. The selected country parameter determines which country gets a black outline and a persistent detail panel under the global map chart legend.

7. Triggers

The Next button triggers the narrative visualization to go to the next scene, and the Back button triggers to go to the previous scene. The year-range control, shown only on Scene 3, is two separate year slider bars, one for the start year and one for the end year. Each slider triggers a data range change for the world map chart. The country dropdown on scene 3 is another trigger that displays/pins the data for that country under the world map chart legend. Clicking a country directly on the map also triggers the country's details to be displayed under the world map chart legend and updates the dropdown country to match, since both are just two different ways of setting the same selected-country parameter. Another trigger is the user's cursor changes to a pointer over any country that has data, this tells the user the countries are clickable.